

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS Document — IEEE 830 / ISO/IEC/IEEE 29148 Format)

SmartCRM AI

AI-Powered Customer Relationship Management (CRM) Platform for Companies & Industries

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Document Version History

This section tracks the revision history of this Software Requirements Specification. It must be updated whenever the document is amended following stakeholder review or scope changes.

Version	Date	Author	Description of Changes	Status
0.1	TBD	Business Analysis Team	Initial draft compiled from project notes, feature list, and workflow descriptions provided by the client.	Draft
1.0	28-Jul-2026	Business Analysis & Solution Architecture Team	Restructured raw project notes into a formal IEEE 830-style SRS; added functional/non-functional requirement statements, user stories, acceptance criteria, and clearly marked assumptions.	Draft for Review

ASSUMPTION: A formal version-control and sign-off log (author names, review dates, approver names) was not present in the source notes. Populate this table as the document proceeds through stakeholder review, and record approvals before development begins.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for SmartCRM AI, an AI-powered Customer Relationship Management platform intended for use by companies and industries to manage customers, leads, sales, enquiries, quotations, communication, service requests, and overall customer relationships.

The purpose of this document is to provide a single, authoritative reference that business stakeholders, project managers, solution architects, developers, quality assurance engineers, and designers can use to understand what the system must do, how it should behave, and the constraints under which it must operate. This document is intended to serve as the primary basis for design, development, testing, and client sign-off.

1.2 Scope

SmartCRM AI is a web-based (and companion mobile) software platform that enables organizations across multiple industries to:

- Maintain a complete, centralized record of customer information (Customer 360°).
- Capture, qualify, and track leads through a defined sales pipeline.
- Manage sales opportunities, quotations, and proposals.
- Communicate with customers through Email, WhatsApp, SMS, and call logging.
- Manage post-sales service requests, complaints, and support tickets.
- Run marketing campaigns and track customer segmentation.
- Track sales team performance, targets, attendance, and commissions.
- View analytics dashboards and generate business reports.
- Leverage Artificial Intelligence (AI) for lead scoring, sales forecasting, recommendations, sentiment analysis, automated content generation, meeting assistance, sales coaching, and duplicate-record detection.

The scope of this release covers the nine core application modules, the nine AI-powered capabilities, a companion mobile application for the sales team, and the supporting technology stack, integrations, and business model as described by the client. Enterprise integrations (ERP, payment gateways), advanced/future features (autonomous AI sales agent, Voice CRM, predictive retention, marketplace integration) are documented separately in Section 23 (Future Enhancements) as they represent a subsequent phase rather than the initial release baseline.

ASSUMPTION: *The source notes did not specify whether this is a single-tenant deployment per client or a true multi-tenant SaaS platform serving many companies simultaneously. Given the stated SaaS subscription model (Starter/Business/Enterprise) and industry-specific CRM variants, this SRS assumes a multi-tenant SaaS architecture where each company/organization operates within its own isolated tenant/workspace. This assumption should be confirmed with stakeholders before architecture finalization.*

1.3 Objectives

The key business and technical objectives of SmartCRM AI, derived from the stated business problems and platform goals, are to:

- 1. Eliminate lost customer enquiries through centralized lead and enquiry capture.
- 2. Improve follow-up discipline through automated reminders and AI-assisted prompts.
- 3. Replace manual, spreadsheet-based sales tracking with a structured digital sales pipeline.
- 4. Provide a complete, unified history of every customer interaction, purchase, and communication.
- 5. Reduce response times to customer enquiries and service requests.
- 6. Improve visibility and management of distributed sales teams.
- 7. Enable data-driven sales forecasting and revenue prediction using AI.
- 8. Surface actionable customer insights (sentiment, buying behavior, churn risk) to sales and support teams.
- 9. Automate repetitive tasks such as quotation generation, email drafting, and meeting summarization.

1.4 Intended Audience

This document is intended for the following audiences, each of whom should read the sections most relevant to their role:

Audience	Primary Sections of Interest
Client / Business Stakeholders	Sections 1, 2, 6, 13, 22, 23, 24, 25
Project Managers	Sections 1, 2, 6, 7, 20, 22, 23, 24
Solution Architects	Sections 2, 8, 9, 10, 18, 20, 21
Backend / Frontend Developers	Sections 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
UI/UX Designers	Sections 11, 12
QA / Test Engineers	Sections 6, 13, 14, 15, 22
DevOps / Infrastructure Engineers	Sections 7, 8, 18, 21
Mobile Application Team	Section 6.10, Section 11, Section 12

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
CRM	Customer Relationship Management — software used to manage a company's interactions with current and prospective customers.
SRS	Software Requirements Specification — this document.
AI	Artificial Intelligence — used across the platform for scoring, prediction, recommendation, and content generation.
NLP	Natural Language Processing — AI technique used for the AI Chat Assistant and Sentiment Analysis features.
LLM	Large Language Model (e.g., OpenAI API models) used to power generative AI features.
Lead	A prospective customer who has shown interest but has not yet been converted into a paying customer.
Opportunity	A qualified sales deal being actively pursued within the Sales Pipeline module.
SLA	Service Level Agreement — a defined time commitment for resolving a service ticket.

Term	Definition
ERP	Enterprise Resource Planning — external business management software integrated with SmartCRM AI.
SaaS	Software as a Service — the subscription-based commercial delivery model for SmartCRM AI.
FR	Functional Requirement — a specific, testable capability the system must provide.
NFR	Non-Functional Requirement — a quality attribute or constraint the system must satisfy.
Customer 360°	A consolidated, single view of all information relating to a customer.
Tenant	An isolated logical instance of the platform data belonging to one client organization (see assumption in Section 1.2).

2. Overall Description

2.1 Product Perspective

SmartCRM AI is a new, independent, AI-augmented CRM product rather than an extension of an existing system. It is designed to serve multiple target industries — manufacturing, engineering, real estate, construction, automobile, industrial equipment supply, software, trading, general services, and healthcare — either as a generic configurable CRM or as an industry-specific variant.

The product is delivered under a SaaS subscription model with three tiers (Starter, Business, Enterprise), industry-specific CRM editions (Manufacturing CRM, Real Estate CRM, Healthcare CRM, Industrial Sales CRM), and a chargeable AI Premium add-on tier (advanced analytics, sales prediction, AI assistant, automation). A companion mobile application ("Sales Team App", built with Flutter) extends core functionality to field sales personnel.

ASSUMPTION: *Precise boundaries between the Starter, Business, and Enterprise plans (feature gating, user limits, pricing) were not specified in the source notes. These must be defined by product/commercial stakeholders and documented in a separate commercial pricing annexure before go-to-market.*

2.2 Product Functions

At a high level, SmartCRM AI performs the following functions, each elaborated as a module in Section 6 (Functional Requirements):

#	Module	Core Function
1	Customer Management (Customer 360°)	Maintain complete, centralized customer information.
2	Lead Management	Capture and manage potential customers through a defined pipeline.
3	Sales Pipeline Management	Track sales opportunities from qualification to closure.
4	Quotation & Proposal Management	Create, price, and track customer quotations and proposals.
5	Communication Management	Unify Email, WhatsApp, SMS, call, and meeting communication.
6	Service & Support Management	Manage complaints, tickets, technician assignment, and SLAs.
7	Marketing Management	Run campaigns, segment customers, and track promotions.
8	Sales Team Management	Track employee performance, targets, attendance, and commissions.
9	Analytics Dashboard & Reporting	Provide dashboards and business reports.
10	AI Capabilities (cross-cutting)	Provide lead scoring, forecasting, recommendations, chat assistance, content generation, sentiment analysis, meeting assistance, sales coaching, and duplicate detection across the above modules.
11	Sales Team Mobile App	Enable field sales staff to log visits, update leads, and receive notifications.

2.3 User Classes and Characteristics

The following user classes have been identified based on the described modules and features. Specific role names and permission boundaries are elaborated as assumptions in Section 14 (Roles & Permissions), since the source notes describe capabilities but do not enumerate formal roles.

User Class	Description	Technical Proficiency
System / Tenant Administrator	Configures the company workspace, manages users, roles, and system settings.	Moderate to high
Sales Manager	Oversees the sales pipeline, sales team performance, targets, and commissions.	Moderate
Sales Executive / Field Salesperson	Manages assigned leads and opportunities, logs visits, uses the mobile app.	Basic to moderate
Marketing Executive	Runs campaigns, manages segmentation and promotions.	Moderate
Customer Support / Service Agent	Manages service tickets, complaints, and technician assignment.	Basic to moderate
Finance / Approval Authority	Approves discounts and reviews quotations and revenue reports.	Moderate
Executive / Business Owner	Views dashboards, analytics, and forecasts for strategic decision-making.	Basic
Customer (indirect user)	Receives communications, quotations, and service updates; does not log into the core system unless a client self-service portal is introduced (see Section 23).	Basic

2.4 Operating Environment

Based on the specified technology stack, SmartCRM AI is expected to operate in the following environment:

- Web application accessible via modern desktop and mobile browsers, built with Next.js, React, and TypeScript, styled using Tailwind CSS.
- Backend services built on Node.js / NestJS and Python FastAPI, exposing REST/JSON APIs (see Section 8).
- Relational data stored in PostgreSQL; non-relational/unstructured/document data stored in MongoDB.
- AI layer powered by the OpenAI API and LangChain orchestration, supplemented by custom Python ML and NLP models.
- Native mobile application for the Sales Team App built with Flutter, targeting Android and iOS.
- Cloud-hosted infrastructure with cloud storage for documents and attachments (see Section 21, Deployment Requirements).

ASSUMPTION: *Specific supported browser versions, minimum OS versions for mobile devices, and cloud provider (AWS/Azure/GCP) were not stated in the source notes. Reasonable industry-standard baselines are proposed in Section 7 (Non-Functional Requirements) and Section 21 (Deployment Requirements) and should be confirmed with the client and infrastructure team.*

2.5 Design and Implementation Constraints

- The system must integrate with third-party services, including WhatsApp Business API, an Email API provider, a Payment Gateway, ERP systems, and Cloud Storage providers — each introducing external dependency, rate limits, and compliance obligations (see Section 19, Integrations).
- The AI layer depends on the OpenAI API (or equivalent LLM provider), introducing dependency on external API availability, cost-per-call, and data-privacy handling for customer information sent to third-party AI services.

- The system must support multiple industry-specific configurations (Manufacturing, Real Estate, Healthcare, Industrial Sales CRM) from a common technical core, implying a configurable/extensible data model rather than hard-coded, industry-specific logic.
- The mobile application must function in low-connectivity field conditions typical of sales visits (see Section 7.4, Reliability, and related assumption).

2.6 Assumptions and Dependencies

The following assumptions and dependencies apply across this SRS. Additional, section-specific assumptions are called out inline throughout the document in highlighted assumption boxes.

10. It is assumed that the client will provide or procure valid commercial API keys/accounts for OpenAI, WhatsApp Business API, the chosen Email API provider, the Payment Gateway, and any ERP system to be integrated.
11. It is assumed that customer data provided to AI features (sentiment analysis, chat assistant, email generation) will be handled in compliance with applicable data protection regulations relevant to the client's operating region(s) (see Section 20, Security Requirements).
12. It is assumed that this is a greenfield (new) build rather than a migration from an existing CRM; no legacy data migration requirements were described in the source notes. If migration is required, a separate Data Migration Plan should be created.
13. It is assumed that the currency of record is Indian Rupees (₹), based on figures used in the source examples (e.g., ₹50 Lakhs, ₹25 Lakhs, ₹5 Crore), with potential for multi-currency support to be confirmed for industries operating internationally.
14. Development will proceed in phases, with the nine core modules and AI features described in Sections 6–7 forming the Minimum Viable Product (MVP)/Phase 1 baseline, and items in Section 23 (Future Enhancements) forming subsequent phases.

6. Functional Requirements

This section specifies the functional requirements of SmartCRM AI, organized by application module. Each module includes: its purpose, a feature list (as provided by the client and expanded into complete requirement statements), one or more representative user stories, and acceptance criteria suitable for QA test-case design. Functional requirements are numbered per module using the pattern FR-<Module Number>.<Sequence> for traceability.

Section 6.10 additionally documents the nine cross-cutting AI capabilities that apply across multiple modules, and Section 6.11 documents the companion Sales Team mobile application.

6.1 Module 1 — Customer Management (Customer 360°)

Purpose: Maintain complete, centralized customer information so that any authorized user can view a single, unified profile of a customer at any time.

Features

- Customer profile management.
- Company details management.
- Contact persons management (multiple contacts per company).
- Industry type classification.
- Purchase history tracking.
- Communication history log.
- Document attachment and storage.

Illustrative Example (from client notes)

Field	Example Value
Company	ABC Engineering Pvt Ltd
Industry	Automobile
Contact	Purchase Manager
Previous Orders	₹50 Lakhs

Functional Requirements

ID	Requirement
FR-1.1	The system shall allow authorized users to create, view, edit, and archive customer records, including company name, industry type, and account status.
FR-1.2	The system shall allow multiple contact persons to be associated with a single company/customer record, each with name, designation, phone, and email.
FR-1.3	The system shall maintain a chronological purchase history for each customer, including order value, product/service, and date.
FR-1.4	The system shall maintain a chronological communication history for each customer, aggregating emails, WhatsApp messages, SMS, and call logs (see Module 5).
FR-1.5	The system shall allow users to upload, view, download, and delete documents (e.g., contracts, purchase orders) attached to a customer record.

ID	Requirement
FR-1.6	The system shall classify each customer by industry type from a configurable list (Manufacturing, Engineering, Real Estate, Construction, Automobile, Industrial Equipment, Software, Trading, Services, Healthcare, and others).
FR-1.7	The system shall present a single consolidated "Customer 360°" view combining profile, contacts, purchase history, communication history, and documents on one screen.

User Story

As a Sales Executive, I want to view a customer's complete profile, purchase history, and communication log in one place, so that I can prepare for a customer interaction without switching between systems.

Acceptance Criteria

- Given a customer record exists, when a user opens the Customer 360° view, then profile details, contact persons, purchase history, communication history, and attached documents are displayed on a single screen.
- Given a user with edit permission, when they update customer details, then the change is saved and reflected immediately with an audit trail entry (see Section 20).
- Given a user uploads a document over the configured size/type limit, then the system rejects the upload and displays a clear validation message (see Section 15).

6.2 Module 2 — Lead Management

Purpose: Capture and manage potential customers from initial interest through to conversion.

Features

- Lead creation.
- Lead source tracking.
- Lead assignment (to sales executives/teams).
- Follow-up reminders.
- Lead status tracking.

Lead Pipeline (as defined by the client)

New Lead → Contacted → Meeting → Quotation → Negotiation → Converted

Functional Requirements

ID	Requirement
FR-2.1	The system shall allow users to create a new lead record capturing, at minimum: lead name, company, contact details, industry, and lead source.
FR-2.2	The system shall track the lead source (e.g., website, referral, exhibition, cold call, marketing campaign) for reporting purposes.
FR-2.3	The system shall allow leads to be assigned to a specific sales executive or team, either manually or via a configurable auto-assignment rule.
FR-2.4	The system shall move a lead through the defined pipeline stages: New Lead, Contacted, Meeting, Quotation, Negotiation, and Converted.

ID	Requirement
FR-2.5	The system shall allow users to set and receive follow-up reminders for each lead, and shall notify the assigned user when a follow-up is due or overdue.
FR-2.6	The system shall allow a converted lead to be transitioned automatically into a Customer record (Module 1) and a Sales Opportunity (Module 3), preserving history.
FR-2.7	The system shall allow users to filter and search leads by status, source, assigned owner, and industry.

User Story

As a Sales Manager, I want to see all leads grouped by pipeline stage, so that I can identify bottlenecks and reassign leads that are not being followed up on time.

Acceptance Criteria

- Given a new enquiry is received, when a user creates a lead, then the lead defaults to "New Lead" status and appears in the assigned executive's queue.
- Given a follow-up date is set on a lead, when that date arrives without the lead being updated, then the system generates a reminder notification (see Section 17).
- Given a lead reaches "Converted" status, when the transition is confirmed, then a linked Customer and Opportunity record are created automatically.

6.3 Module 3 — Sales Pipeline Management

Purpose: Track sales opportunities from qualification through to closure, giving sales managers visibility into deal value and stage.

Features

- Opportunity tracking.
- Sales stage tracking.
- Deal value tracking.
- Expected closing date tracking.
- Sales team performance visibility.

Illustrative Example (from client notes)

Field	Example Value
Customer	XYZ Industries
Product	Industrial Crane
Value	₹25 Lakhs
Stage	Negotiation

Functional Requirements

ID	Requirement
FR-3.1	The system shall allow users to create a Sales Opportunity linked to a customer, recording product/service, deal value, and expected closing date.
FR-3.2	The system shall track each opportunity through configurable sales stages (e.g., Qualification, Proposal, Negotiation, Won, Lost).
FR-3.3	The system shall display opportunities in a pipeline/kanban view grouped by stage, showing cumulative deal value per stage.
FR-3.4	The system shall record the reason when an opportunity is marked "Lost", to support future analysis.
FR-3.5	The system shall provide sales-team-level and individual-level performance views summarizing open, won, and lost opportunity value.

User Story

As a Sales Manager, I want to view all open opportunities and their stage and value, so that I can forecast likely revenue for the current period.

Acceptance Criteria

- Given an opportunity is created, when a user changes its stage, then the pipeline view updates in real time and the stage-change is timestamped and attributed to the user.
- Given an opportunity is marked "Lost", then the system requires a loss reason before the record can be closed.

6.4 Module 4 — Quotation & Proposal Management

Purpose: Enable sales staff to generate accurate, professional quotations and proposals quickly, with AI assistance for pricing and profitability analysis.

Features

- Quotation creation.
- Product pricing management.
- Discount approval workflow.
- Generation of PDF proposals.
- Customer approval tracking.

AI Features

- Automatic quotation generation.
- Price recommendation.
- Profit margin analysis.

Functional Requirements

ID	Requirement
FR-4.1	The system shall allow users to create a quotation linked to a customer and/or opportunity, listing line items, quantities, unit prices, and discounts.
FR-4.2	The system shall maintain a product/service price list used to populate quotation line items.

ID	Requirement
FR-4.3	The system shall route discounts above a configurable threshold to a designated approver before the quotation can be finalized (see Section 13, Business Rules).
FR-4.4	The system shall generate a downloadable, professionally formatted PDF proposal document from a quotation.
FR-4.5	The system shall track the customer's response to a quotation (Pending, Viewed, Approved, Rejected, Expired).
FR-4.6 (AI)	The system shall provide an AI-assisted quotation draft based on the customer profile, product selection, and historical pricing, which the user can review and edit before sending.
FR-4.7 (AI)	The system shall provide an AI-generated price recommendation for a given product/customer combination, based on historical pricing and win/loss data.
FR-4.8 (AI)	The system shall calculate and display an estimated profit margin for a quotation before it is sent to the customer.

User Story

As a Sales Executive, I want the system to draft a quotation automatically from an opportunity, so that I can send an accurate proposal to the customer within minutes instead of hours.

Acceptance Criteria

- Given a discount exceeds the configured approval threshold, when the user attempts to send the quotation, then the system blocks sending until an approver confirms the discount.
- Given a quotation is finalized, when the user selects "Generate PDF", then a correctly formatted proposal document is produced containing all line items, totals, and company branding.

ASSUMPTION: Discount approval thresholds, approver hierarchy, and price-list ownership were not specified in the source notes; default values must be configured per client during implementation (see Section 13, Business Rules).

6.5 Module 5 — Communication Management

Purpose: Unify all customer communication channels within the CRM so that no interaction history is lost across systems.

Features

- Email integration.
- WhatsApp communication.
- SMS alerts.
- Call records.
- Meeting scheduling.

Functional Requirements

ID	Requirement
FR-5.1	The system shall integrate with an Email API to send and receive customer emails from within the CRM, logging each email against the relevant customer/lead record.

ID	Requirement
FR-5.2	The system shall integrate with the WhatsApp Business API to send and receive WhatsApp messages, logging each message against the relevant customer/lead record.
FR-5.3	The system shall send SMS alerts to customers or internal users for configurable events (e.g., quotation sent, follow-up due, ticket update).
FR-5.4	The system shall allow users to log call records manually (or automatically, where telephony integration exists), including call outcome and notes.
FR-5.5	The system shall allow users to schedule meetings with customers, with optional reminders and calendar integration.
FR-5.6	The system shall present all communication types (email, WhatsApp, SMS, call, meeting) in a single chronological timeline on the customer record.

User Story

As a Sales Executive, I want to see every email, WhatsApp message, call, and meeting with a customer in one timeline, so that I never lose context between interactions.

Acceptance Criteria

- Given a WhatsApp message is sent to a customer through the CRM, then the message and any reply appear in the customer's communication timeline within a reasonable delay (see Section 7.1, Performance).
- Given an SMS alert event occurs (e.g., quotation sent), then the configured recipient receives the SMS and the send status is logged.

6.6 Module 6 — Service & Support Management

Purpose: Manage post-sales customer complaints and service requests to ensure timely resolution and customer satisfaction.

Features

- Customer complaint logging.
- Service ticket management.
- Technician assignment.
- SLA (Service Level Agreement) tracking.
- Feedback management.

Illustrative Example (from client notes)

Field	Example Value
Issue	Machine Not Working
Priority	High
Assigned	Service Engineer

Functional Requirements

ID	Requirement
FR-6.1	The system shall allow customers or internal users to log a complaint or service request against a customer/asset, capturing issue description and priority (e.g., Low, Medium, High, Critical).
FR-6.2	The system shall generate a unique service ticket for each logged complaint/request and track its status (e.g., Open, Assigned, In Progress, Resolved, Closed).
FR-6.3	The system shall allow tickets to be assigned to a technician or service engineer, either manually or through configurable auto-assignment rules.
FR-6.4	The system shall track SLA timers per ticket priority and shall flag or escalate tickets that are at risk of, or have breached, their SLA.
FR-6.5	The system shall allow customers to provide feedback/rating upon ticket closure, and shall store this feedback against the customer and technician records.

User Story

As a Service Manager, I want High priority tickets to be flagged automatically when they approach their SLA deadline, so that I can intervene before the SLA is breached.

Acceptance Criteria

- Given a High priority ticket is created, when the configured SLA time threshold is reached without resolution, then the system raises an escalation notification to the Service Manager.
- Given a ticket is marked "Resolved", when the customer submits feedback, then the feedback score is recorded and reflected in service reports (see Section 18).

ASSUMPTION: Specific SLA durations per priority level were not provided in the source notes; these must be defined per client/industry during configuration.

6.7 Module 7 — Marketing Management

Purpose: Enable marketing teams to run targeted campaigns and track their effectiveness, enhanced by AI-driven targeting and personalization.

Features

- Campaign management.
- Email marketing.
- Customer segmentation.
- Promotion tracking.

AI Features

- Customer targeting.
- Campaign prediction.
- Personalized offers.

Functional Requirements

ID	Requirement
FR-7.1	The system shall allow marketing users to create, schedule, and launch campaigns across email and other supported channels.
FR-7.2	The system shall allow customers to be segmented by attributes such as industry, purchase history, engagement level, and lead source.
FR-7.3	The system shall track promotion/offer usage and redemption against customer records.
FR-7.4	The system shall report on campaign performance metrics (e.g., sent, opened, clicked, converted).
FR-7.5 (AI)	The system shall use AI to recommend target customer segments most likely to respond to a given campaign.
FR-7.6 (AI)	The system shall use AI to predict the likely performance (e.g., expected response/conversion rate) of a planned campaign before launch.
FR-7.7 (AI)	The system shall use AI to generate personalized offer recommendations for individual customers based on their profile and purchase history.

User Story

As a Marketing Executive, I want the system to recommend which customer segment to target for a new promotion, so that my campaign achieves a higher conversion rate with less manual analysis.

Acceptance Criteria

- Given a campaign is created, when the user requests an AI targeting recommendation, then the system returns a suggested customer segment with a stated rationale.
- Given a campaign has concluded, then campaign performance metrics are available in the Reporting module (Section 18).

6.8 Module 8 — Sales Team Management

Purpose: Track the performance, activity, attendance, and compensation of the sales team.

Features

- Employee performance tracking.
- Target tracking.
- Sales activity tracking.
- Attendance tracking.
- Commission calculation.

Functional Requirements

ID	Requirement
FR-8.1	The system shall allow managers to set sales targets (revenue and/or unit-based) per salesperson or team, for a defined period.
FR-8.2	The system shall track actual sales performance against target and display progress (e.g., percentage attainment).
FR-8.3	The system shall log sales activities (calls, visits, meetings, quotations sent) per salesperson for activity reporting.
FR-8.4	The system shall track sales team attendance, including check-in/check-out where captured via the mobile application (see Section 6.11).
FR-8.5	The system shall calculate sales commission based on configurable commission rules (e.g., percentage of closed deal value).

User Story

As a Sales Manager, I want to see each salesperson's target attainment and activity level side by side, so that I can identify who needs coaching or support.

Acceptance Criteria

- Given a sales target is set for a salesperson, when opportunities are won during the target period, then attainment percentage updates automatically.
- Given commission rules are configured, when a deal is marked Won, then the corresponding commission is calculated and recorded for payroll reference.

ASSUMPTION: Commission calculation formulas and payroll system integration were not detailed in the source notes; this SRS assumes commission figures are calculated within SmartCRM AI for reference/reporting and may require export to, or integration with, an external payroll/HR system — to be confirmed.

6.9 Module 9 — Analytics Dashboard

Purpose: Provide business owners and managers with a real-time, consolidated view of key CRM metrics.

Illustrative Example (from client notes)

Metric	Example Value
Total Leads	2,500
Converted	450
Revenue	₹5 Crore

Metric	Example Value
Growth	22%

Reports

- Sales reports.
- Customer reports.
- Revenue reports.
- Team performance reports.

Functional Requirements

ID	Requirement
FR-9.1	The system shall display a configurable dashboard summarizing total leads, converted leads, conversion rate, total revenue, and period-over-period growth.
FR-9.2	The system shall provide a Sales Report summarizing opportunities by stage, value, and owner over a selectable date range.
FR-9.3	The system shall provide a Customer Report summarizing customer counts, segments, and activity over a selectable date range.
FR-9.4	The system shall provide a Revenue Report summarizing invoiced/closed revenue by period, product, industry, and salesperson.
FR-9.5	The system shall provide a Team Performance Report summarizing target attainment, activity, and commission by salesperson/team.
FR-9.6	The system shall allow dashboards and reports to be filtered by date range, industry, team, and individual user, where applicable to the user's role permissions.
FR-9.7	The system shall allow reports to be exported (see Section 18, Reporting Requirements).

User Story

As a Business Owner, I want to open a single dashboard showing total leads, conversions, revenue, and growth, so that I can assess business health at a glance without requesting reports manually.

Acceptance Criteria

- Given the dashboard is loaded, then all headline metrics reflect data as of the most recent successful data refresh, with the refresh time displayed to the user.
- Given a user without cross-team visibility permission, when they view the dashboard, then figures are scoped to their own team/records only (see Section 14, Roles & Permissions).

6.10 Cross-Cutting AI Capabilities

SmartCRM AI embeds nine distinct AI-powered capabilities that operate across the modules described above. Each is specified below with its purpose, illustrative example (as supplied by the client), and functional requirements.

6.10.1 AI Feature 1 — AI Lead Scoring

Purpose: Identify high-potential leads so sales effort is prioritized effectively.

The AI analyzes: industry, budget, previous interactions, buying behavior, and engagement.

Field	Example Value
Lead	ABC Industries
AI Score	92%
Probability	High chance of purchase
ID	Requirement
FR-AI-1.1	The system shall calculate an AI Lead Score (0–100%) for each lead based on industry, budget, historical interactions, buying behavior, and engagement level.
FR-AI-1.2	The system shall display the lead score and a qualitative purchase-probability label (e.g., Low/Medium/High) on the lead record.
FR-AI-1.3	The system shall allow leads to be sorted and filtered by AI score to help prioritize follow-up.

6.10.2 AI Feature 2 — AI Sales Prediction

The AI predicts: future sales, revenue, deal probability, and monthly targets.

Field	Example Value
Next Month Forecast — Expected Revenue	₹1.2 Crore
Growth	18%
ID	Requirement
FR-AI-2.1	The system shall generate a forecasted revenue figure for a selectable upcoming period (e.g., next month/quarter) based on historical sales data and current pipeline.
FR-AI-2.2	The system shall estimate the probability of closing each open opportunity based on historical conversion patterns.
FR-AI-2.3	The system shall compare the forecast against the configured sales target and display the projected growth/variance percentage.

6.10.3 AI Feature 3 — AI Customer Recommendation

The AI suggests: which product to offer, when to contact the customer, and the best sales approach.

Field	Example Value
Customer bought	Industrial Motor
Recommendation	Offer: Maintenance Contract
ID	Requirement

Field	Example Value
FR-AI-3.1	The system shall recommend complementary or follow-up products/services for a customer based on their purchase history and profile.
FR-AI-3.2	The system shall recommend an optimal time window to contact a given customer based on historical engagement patterns.
FR-AI-3.3	The system shall suggest a recommended sales approach or talking points for a given customer/opportunity.

6.10.4 AI Feature 4 — AI Chat Assistant

Employees can ask natural-language questions (e.g., "Show customers who haven't purchased in 6 months") and receive an answer with a recommended action.

Field	Example Value
Query	Show customers who haven't purchased in 6 months
Result	Found: 125 Customers
Recommended Action	Send reactivation campaign
ID	Requirement
FR-AI-4.1	The system shall provide a natural-language chat interface allowing authorized users to query CRM data (e.g., customers, leads, sales) using plain-language questions.
FR-AI-4.2	The system shall return query results as structured data (e.g., a filtered customer list) accompanied by a plain-language summary.
FR-AI-4.3	The system shall suggest a recommended next action related to the query result, where applicable (e.g., launching a reactivation campaign).
FR-AI-4.4	The system shall restrict chat-assistant query results to data the requesting user is authorized to view (see Section 14, Roles & Permissions).

6.10.5 AI Feature 5 — AI Email & Message Generator

The AI creates: sales emails, follow-up messages, marketing campaign content, and proposal content.

Field	Example Value
Input	Follow up with customer about quotation
Output	Professional email generated automatically
ID	Requirement
FR-AI-5.1	The system shall generate a draft email or message from a short user instruction (e.g., "follow up on quotation"), pre-filled with relevant customer/deal context.
FR-AI-5.2	The system shall allow the user to review, edit, and approve AI-generated content before it is sent to a customer.
FR-AI-5.3	The system shall support AI-generated content for sales emails, follow-up messages, marketing campaign copy, and proposal narrative text.

6.10.6 AI Feature 6 — AI Customer Sentiment Analysis

The AI analyzes emails, reviews, and chat messages, and detects happy customers, angry customers, and risk of losing customers.

ID	Requirement
FR-AI-6.1	The system shall analyze the sentiment of customer emails, reviews, and chat/WhatsApp messages, classifying each as Positive, Neutral, or Negative.
FR-AI-6.2	The system shall flag customers exhibiting a pattern of negative sentiment as at risk of churn, and surface this flag to the assigned account owner and manager.
FR-AI-6.3	The system shall display an aggregated sentiment trend per customer over time.

6.10.7 AI Feature 7 — AI Meeting Assistant

Features: meeting recording, conversation summary, action items, and follow-up reminders.

ID	Requirement
FR-AI-7.1	The system shall support recording of customer meetings (audio, subject to consent and applicable law — see Section 20).
FR-AI-7.2	The system shall generate an AI summary of the meeting conversation.
FR-AI-7.3	The system shall extract action items from the meeting and allow them to be converted into follow-up tasks/reminders.

ASSUMPTION:
Meeting recording raises consent and data-privacy obligations that vary by jurisdiction. The source notes do not specify a consent-capture mechanism; this SRS assumes the system will require explicit participant consent before recording begins, to be confirmed with legal/compliance stakeholders.

6.10.8 AI Feature 8 — AI Sales Coach

The AI analyzes salesperson performance and provides coaching suggestions.

Field	Example Value
Meetings	50
Conversion	10%

Field		Example Value
Suggestion		Improve follow-up frequency
ID	Requirement	
FR-AI-8.1	The system shall analyze each salesperson's activity and conversion metrics (meetings held, conversion rate, follow-up frequency, etc.).	
FR-AI-8.2	The system shall generate personalized, actionable coaching suggestions for each salesperson based on their performance pattern.	
FR-AI-8.3	The system shall make coaching suggestions visible to the salesperson and their manager.	

6.10.9 AI Feature 9 — AI Duplicate Customer Detection

The AI detects duplicate companies, duplicate contacts, and incorrect data.

ID	Requirement
FR-AI-9.1	The system shall detect probable duplicate company/customer records based on name similarity, contact details, and other identifying attributes.
FR-AI-9.2	The system shall detect probable duplicate contact-person records within or across customer accounts.
FR-AI-9.3	The system shall flag records with likely incorrect or incomplete data (e.g., invalid email/phone formats) for user review.
FR-AI-9.4	The system shall provide a guided merge workflow allowing an authorized user to review and merge duplicate records without losing history.

6.11 Sales Team Mobile Application

Purpose: Extend core CRM functionality to field sales personnel via a native mobile application, built using Flutter, targeting Android and iOS.

Features

- Customer visit logging.
- Lead updates from the field.
- Meeting reports.
- Push notifications.

Functional Requirements

ID	Requirement
FR-M.1	The mobile application shall allow a salesperson to log a customer visit, including date/time, location, purpose, and outcome notes.
FR-M.2	The mobile application shall allow a salesperson to view and update assigned leads and opportunities, including changing pipeline stage.
FR-M.3	The mobile application shall allow a salesperson to submit a meeting report, including summary notes and next steps.
FR-M.4	The mobile application shall receive push notifications for assigned leads, follow-up reminders, and manager messages.
FR-M.5	The mobile application shall support attendance check-in/check-out, feeding into Module 8 (Sales Team Management).
FR-M.6	The mobile application shall allow core functions (viewing assigned leads, drafting visit notes) to remain usable with intermittent connectivity, synchronizing data once connectivity is restored.

User Story

As a Field Salesperson, I want to log a customer visit and update the lead stage from my phone immediately after a meeting, so that the CRM stays current without requiring me to return to a desktop.

Acceptance Criteria

- Given the salesperson is offline, when they create a visit log entry, then the entry is stored locally and synchronized automatically once network connectivity is available.
- Given a lead is reassigned to a salesperson, then that salesperson receives a push notification within a reasonable delay (see Section 7.1, Performance).

ASSUMPTION: *Offline-first behavior and GPS-based visit verification were not explicitly described in the source notes but are inferred as standard expectations for a field-sales mobile CRM app; scope and precision requirements (e.g., geofencing) should be confirmed with stakeholders.*

7. Non-Functional Requirements

The source notes focused primarily on functional features; specific non-functional targets (response times, uptime percentages, etc.) were not stated. The requirements below reflect industry-standard baselines appropriate for a multi-tenant SaaS CRM handling customer and sales data, and are marked as assumptions where a numeric target has been proposed rather than confirmed by the client.

7.1 Performance

- The system shall load the primary dashboard within 3 seconds under normal load conditions.
- The system shall return search/filter results (e.g., customer, lead search) within 2 seconds for datasets up to the agreed scale target.
- AI-powered features (lead scoring, recommendations) that do not require a live LLM call shall return results within 2 seconds; generative AI features (email drafting, chat assistant) that call an external LLM shall return results within 8 seconds under normal conditions.

ASSUMPTION: *Numeric performance targets above are proposed industry-standard baselines, not figures supplied by the client. They should be validated against expected data volumes and concurrent user counts.*

7.2 Security

- All data in transit shall be encrypted using TLS 1.2 or higher.
- All sensitive data at rest (passwords, payment details, personal data) shall be encrypted.
- The system shall enforce role-based access control (RBAC) as defined in Section 14.
- The system shall maintain an audit trail of create/update/delete actions on customer, financial, and user-management records.
- See Section 20 for full Security Requirements.

7.3 Scalability

- The system architecture shall support horizontal scaling of backend services to accommodate growth in tenants, users, and data volume.
- The system shall support the addition of new tenants (client companies) without downtime to existing tenants.

7.4 Reliability

- The system shall implement retry/queueing logic for outbound integrations (WhatsApp, Email, SMS, Payment Gateway) to handle transient third-party failures without data loss.
- The mobile application shall queue field-captured data locally when offline and synchronize automatically upon reconnection (see FR-M.6).

7.5 Availability

- The production system shall target a minimum of 99.5% monthly uptime, excluding scheduled maintenance windows communicated in advance.

ASSUMPTION: *The 99.5% uptime figure is a proposed baseline; the client's actual SLA commitment (if any, particularly for Enterprise-tier subscribers) should be confirmed and may need to be higher.*

7.6 Maintainability

- The codebase shall follow modular architecture aligned to the modules defined in Section 6, to allow independent development, testing, and deployment of features.
- Configuration such as pipeline stages, SLA thresholds, and commission rules shall be externally configurable rather than hard-coded, to support multiple industries and clients.

7.7 Accessibility

- The web application shall aim to meet WCAG 2.1 Level AA accessibility guidelines for key workflows (forms, dashboards, navigation).

ASSUMPTION: *No specific accessibility target was provided in the source notes; WCAG 2.1 AA is proposed as an industry-standard baseline for enterprise SaaS products and should be confirmed with the client.*

7.8 Browser Compatibility

- The web application shall support the latest two major versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

7.9 Mobile Responsiveness

- The web application shall be fully responsive and usable on tablet and mobile browser viewports for core workflows (lead view, customer view, notifications), in addition to the dedicated Flutter mobile app for field sales use (Section 6.11).

8. System Architecture Overview

SmartCRM AI follows a layered, service-oriented architecture consistent with the technology stack specified by the client.

Layer	Technology	Responsibility
Presentation (Web)	Next.js, React, TypeScript, Tailwind CSS	Renders the web application UI for desktop and responsive mobile browsers.
Presentation (Mobile)	Flutter (Android & iOS)	Delivers the Sales Team App for field personnel.
Application / API	Node.js / NestJS, Python FastAPI	Implements business logic and exposes REST/JSON APIs consumed by the web and mobile clients.
AI Layer	OpenAI API, LangChain, Python ML models, NLP models	Powers lead scoring, prediction, recommendations, chat assistant, content generation, sentiment analysis, and coaching features.
Data Layer	PostgreSQL, MongoDB	PostgreSQL stores structured, relational data (customers, opportunities, tickets, users). MongoDB stores unstructured/flexible data such as chat logs, AI outputs, and activity streams.
Integration Layer	WhatsApp API, Email API, Payment Gateway, ERP Integration, Cloud Storage	Connects SmartCRM AI to external systems and services (see Section 19).

High-level data/control flow: client applications (web and mobile) communicate with the Application/API layer over authenticated HTTPS REST APIs; the Application layer reads/writes to PostgreSQL and MongoDB, invokes the AI layer for AI-powered features, and dispatches outbound events (email, WhatsApp, SMS, notifications) via the Integration layer.

ASSUMPTION: *The precise split of responsibilities between the Node.js/NestJS service and the Python FastAPI service was not detailed in the source notes. This SRS assumes NestJS handles core transactional CRM business logic (customers, leads, opportunities, tickets) while FastAPI hosts AI/ML-oriented endpoints (scoring, prediction, NLP), given Python's ecosystem strength for AI workloads. This split should be validated by the architecture team.*

9. Database Requirements

The source notes specify the database technologies (PostgreSQL and MongoDB) but not a detailed schema. This section defines the database requirements and proposes a high-level, non-exhaustive entity list derived from the functional requirements in Section 6. A full data dictionary and Entity-Relationship Diagram (ERD) should be produced during the technical design phase.

9.1 Database Technology Requirements

- PostgreSQL shall be used as the system of record for structured, relational, transactional data requiring strong consistency (customers, contacts, leads, opportunities, quotations, tickets, users, roles, targets, commissions).
- MongoDB shall be used for semi-structured or high-volume data such as communication logs, AI feature outputs (scores, sentiment results, chat transcripts), and activity/audit streams.
- All tenant data shall be logically isolated to prevent cross-tenant data access (see Section 20, Security Requirements, and the multi-tenancy assumption in Section 1.2).
- Database backups shall be performed on a defined schedule with a documented retention policy (see Section 21, Deployment Requirements).

9.2 High-Level Entities (Indicative)

Entity	Description	Related Module(s)
Tenant / Organization	The client company using the platform.	All
User	Internal system user with a role.	All
Customer / Company	Customer 360° master record.	Module 1
Contact	Individual contact person linked to a Customer.	Module 1
Lead	Prospective customer in the sales funnel.	Module 2
Opportunity	Sales deal in progress.	Module 3
Quotation	Pricing document generated for a customer/opportunity.	Module 4
Product / Price List Item	Sellable product or service with pricing.	Module 4
Communication Log	Email/WhatsApp/SMS/Call/Meeting record.	Module 5
Service Ticket	Complaint or support request record.	Module 6
Campaign	Marketing campaign definition and results.	Module 7
Sales Target	Target definition per user/team/period.	Module 8
Attendance Record	Check-in/check-out log.	Module 8

Entity	Description	Related Module(s)
Commission Record	Calculated commission per closed deal.	Module 8
AI Insight	Stored AI output (score, prediction, sentiment, recommendation).	AI Features
Document / Attachment	File metadata and cloud storage reference.	Module 1, 4, 6
Audit Log	Record of create/update/delete actions.	Cross-cutting
ASSUMPTION: <i>This entity list is indicative and derived from the functional requirements; it is not an exhaustive data model. Field-level attributes, keys, and relationships must be finalized in a formal ERD/data dictionary during technical design.</i>		

10. API Requirements

The source notes did not include an API specification. Given the Next.js/React web client, Flutter mobile client, and NestJS/FastAPI backend, this SRS assumes a REST/JSON API architecture and defines baseline API requirements. A full OpenAPI/Swagger specification should be produced during technical design.

10.1 General API Requirements

- All client-facing functionality (web and mobile) shall be served through versioned REST APIs (e.g., /api/v1/...).
- APIs shall authenticate requests using token-based authentication (e.g., JWT/OAuth2) and shall reject unauthenticated requests.
- APIs shall enforce role-based authorization consistent with Section 14 (Roles & Permissions) on every endpoint.
- APIs shall return standardized JSON error responses including an error code, human-readable message, and correlation/request ID (see Section 16, Error Handling).
- APIs that trigger AI processing (Section 6.10) shall support asynchronous handling (e.g., polling or webhook callback) for operations that may exceed typical synchronous response times.

10.2 Indicative API Groups

API Group	Example Endpoints (Indicative)	Supports Module
Authentication	/auth/login, /auth/refresh, /auth/logout	Cross-cutting
Customers	/customers, /customers/{id}, /customers/{id}/history	Module 1
Leads	/leads, /leads/{id}/status	Module 2
Opportunities	/opportunities, /opportunities/{id}/stage	Module 3
Quotations	/quotations, /quotations/{id}/pdf	Module 4
Communications	/communications/email, /communications/whatsapp, /communications/sms	Module 5
Service Tickets	/tickets, /tickets/{id}/assign	Module 6
Campaigns	/campaigns, /campaigns/{id}/performance	Module 7
Sales Team	/targets, /attendance, /commissions	Module 8
Dashboard & Reports	/dashboard, /reports/sales, /reports/revenue	Module 9
AI Services	/ai/lead-score, /ai/forecast, /ai/chat, /ai/generate-email, /ai/sentiment	AI Features
Integrations	/integrations/erp, /integrations/payment, /integrations/whatsapp-webhook	Section 19

ASSUMPTION:
*Endpoint names
and payload
structures above
are illustrative*

API Group	Example Endpoints (Indicative)	Supports Module
<i>examples for planning purposes only and are not a binding API contract. The engineering team shall author a definitive OpenAPI/Swagger specification during the design phase.</i>		

11. UI/UX Requirements

The source document did not include wireframes, mockups, or screenshots; it described features and example data structures only. This section defines UI/UX requirements at a principle level, to be elaborated into detailed wireframes and a design system during the UI/UX design phase.

11.1 General UI/UX Principles

- The interface shall present a persistent primary navigation (e.g., left sidebar) covering: Dashboard, Customers, Leads, Opportunities, Quotations, Communications, Service Tickets, Campaigns, Sales Team, Reports, and AI Assistant.
- The Customer 360° view (FR-1.7) shall be presented as a single-page, tabbed or scrollable layout combining profile, contacts, history, and documents, minimizing navigation between screens.
- The Lead and Opportunity pipelines shall be presented using a Kanban-style board with drag-and-drop stage transitions, reflecting the pipeline defined in Section 6.2 and 6.3.
- AI-generated content (quotation drafts, emails, recommendations) shall be visually distinguished from user-authored content and shall always require explicit user review/approval before being sent externally.
- The design shall follow a consistent design system (typography, color palette, spacing, and component library) across web and mobile surfaces.
- Forms shall provide inline validation feedback consistent with Section 15 (Validation Rules).

11.2 Dashboard & Reporting UI

- The Analytics Dashboard (Module 9) shall present headline metrics as summary cards (Total Leads, Converted, Revenue, Growth) followed by detailed charts and report links, mirroring the example structure provided by the client.
- Charts shall support date-range filtering and, where role-permitted, drill-down into underlying records.

ASSUMPTION: No brand guidelines, color palette, or logo were provided in the source document. Visual design (colors, typography, branding) must be finalized in collaboration with the client's design/brand team.

12. Navigation Flow

The following high-level navigation flow reflects the lead-to-cash and service lifecycle described in the client notes.

12.1 Primary Lead-to-Customer Flow

New Lead → Contacted → Meeting → Quotation → Negotiation → Converted → Customer (Module 1) + Opportunity Won (Module 3)

12.2 Quotation Flow

Opportunity → Create Quotation → (AI-Assisted Draft) → Pricing & Discount → [Discount Approval if threshold exceeded] → Generate PDF → Send to Customer → Track Response (Pending/Viewed/Approved/Rejected/Expired)

12.3 Service Ticket Flow

Complaint/Request Logged → Ticket Created (Open) → Technician Assigned → In Progress → [SLA Monitoring] → Resolved → Customer Feedback → Closed

12.4 Navigation Map (Primary Menu → Sub-Screens)

Primary Menu Item	Sub-Screens
Dashboard	Overview, Sales Report, Customer Report, Revenue Report, Team Performance
Customers	Customer List, Customer 360° Detail, Documents, Communication Timeline
Leads	Lead Pipeline (Kanban), Lead Detail, Lead Source Report
Opportunities	Opportunity Pipeline (Kanban), Opportunity Detail
Quotations	Quotation List, Create/Edit Quotation, PDF Preview
Communications	Unified Inbox (Email/WhatsApp/SMS), Call Log, Meeting Scheduler
Service Tickets	Ticket List, Ticket Detail, SLA Dashboard
Campaigns	Campaign List, Create Campaign, Segmentation, Performance Report
Sales Team	Targets, Attendance, Commission, Activity Log
AI Assistant	Chat Interface, Insights (Lead Score, Forecast, Sentiment, Coaching)
Settings / Admin	Users & Roles, Pipeline Configuration, SLA Configuration, Integrations

13. Business Rules

Business rules extracted or reasonably inferred from the client notes are listed below. Rules marked as assumptions require confirmation with business stakeholders and configuration per client/industry.

ID	Business Rule	Source
BR-1	A lead must progress through the defined pipeline stages in order: New Lead → Contacted → Meeting → Quotation → Negotiation → Converted (stages may be skipped only via explicit user override, which shall be logged).	Client notes (Lead Pipeline)
BR-2	A converted lead automatically generates a linked Customer record and Sales Opportunity.	Inferred from Lead/Customer/Opportunity relationship
BR-3	Discounts on a quotation above a configurable threshold require approval before the quotation can be sent to the customer.	Client notes ("Discount approval")
BR-4	Service tickets are assigned a priority level, and each priority level has an associated SLA timer that triggers escalation if breached.	Client notes (SLA tracking)
BR-5	Sales commission is calculated based on closed (Won) opportunity value using configurable commission rules.	Client notes (Commission calculation)
BR-6	AI-generated content (quotations, emails, campaign copy) must be reviewed and explicitly approved by a human user before being sent to a customer.	Assumption — standard AI governance practice
BR-7	Customer, lead, and opportunity data is scoped to the owning tenant (client organization) and is not visible across tenants.	Assumption — required for multi-tenant SaaS model
BR-8	A customer record may not be permanently deleted if it has associated financial history (quotations, orders); it may only be archived/deactivated.	Assumption — standard data-integrity practice

14. Roles & Permissions

The client notes described user activities (e.g., sales managers, sales executives, service engineers, marketing users) but did not define a formal role/permission matrix. The following role structure is proposed based on the described user classes (Section 2.3) and should be validated and refined with the client.

Role	Customers	Leads/Opportunities	Quotations	Service Tickets	Reports/Dashboard	Admin Settings
Tenant Administrator	Full	Full	Full	Full	Full (all teams)	Full
Sales Manager	Full (team)	Full (team)	Approve/Full	View	Full (team)	Limited
Sales Executive	View/Edit (own)	Full (own)	Create/Edit (own)	Create	View (own)	None
Marketing Executive	View	View (leads only)	None	None	Campaign reports	Limited (campaigns)
Service Agent	View	None	None	Full (assigned)	Service reports	None
Finance/Approver	View	View	Approve discounts	None	Revenue reports	None
Executive/Owner	View	View	View	View	Full	None

ASSUMPTION: *The role matrix above is a proposed baseline derived from the described user activities in the source notes; it is not an explicit client specification. It must be reviewed, adjusted, and formally signed off with the client — including confirmation of whether permissions should be further customizable per tenant.*

15. Validation Rules

The source notes did not specify field-level validation rules. The following baseline validation rules are proposed based on standard CRM data-entry practices and the data fields implied by the functional requirements in Section 6.

Field / Area	Validation Rule
Email address	Must conform to standard email format; duplicate emails within a tenant flagged per FR-AI-9.3.
Phone number	Must conform to a valid phone number format for the configured country/region.
Customer / Company name	Required field; minimum length enforced; checked for probable duplicates (FR-AI-9.1).
Quotation line items	Quantity and unit price must be positive numeric values; discount percentage must be between 0–100%.
Discount threshold	Discounts exceeding the configured approval threshold are blocked from sending until approved (BR-3).
Document uploads	Restricted to approved file types (e.g., PDF, DOCX, XLSX, JPG, PNG) and a maximum file size (e.g., 25 MB), to be confirmed.
Lead/Opportunity stage transitions	Must follow the configured pipeline sequence, with overrides logged (BR-1).
Mandatory fields	Required fields (e.g., customer name, lead source, ticket priority) must be completed before a record can be saved as "active".

ASSUMPTION:
Specific validation thresholds (file size limits, phone number formats by country, etc.) were not provided in the source notes and are proposed as reasonable defaults for confirmation.

16. Error Handling

The source notes did not specify error-handling behavior. The following baseline requirements are proposed to ensure a consistent, professional error experience across the platform.

- The system shall display clear, human-readable error messages for validation failures, distinguishing them from system/technical errors.
- The system shall not expose internal technical details (stack traces, database errors) to end users; such details shall be logged server-side with a correlation/request ID shown to the user for support purposes.
- The system shall handle third-party integration failures (WhatsApp, Email, SMS, Payment Gateway, ERP, OpenAI API) gracefully, informing the user that the action could not be completed and, where applicable, queueing the action for retry (see Section 7.4, Reliability).
- The mobile application shall clearly indicate offline status and inform the user which actions are queued for synchronization.
- Critical system errors (e.g., failed AI service calls, integration outages) shall be logged and, where configured, trigger an alert to the technical operations team.

17. Notifications

The client notes reference Email, WhatsApp, and SMS as communication channels (Module 5), and the mobile app requires push notifications (Section 6.11). This section consolidates notification requirements across the platform.

Channel	Trigger Events (Indicative)	Recipient
Email	Quotation sent, campaign emails, follow-up reminders, ticket updates, weekly report digest.	Customer / Internal user
WhatsApp	Order/quotation updates, service ticket status, promotional messages (subject to opt-in/consent).	Customer
SMS	Time-sensitive alerts: quotation sent, follow-up due, ticket escalation.	Customer / Internal user
Push (Mobile App)	Lead assignment, follow-up due, manager message, ticket assignment.	Internal user (field sales/service)
In-app Notification	SLA breach warnings, discount approval requests, AI insight alerts (e.g., churn risk).	Internal user

17.1 Functional Requirements

- The system shall allow tenant administrators to configure which events trigger notifications and through which channel(s).
- The system shall respect customer communication preferences and opt-out/consent status for marketing messages (WhatsApp, Email, SMS), consistent with applicable regulations.
- The system shall log the delivery status (sent, delivered, failed) of each notification for audit and troubleshooting.

ASSUMPTION: *Specific default trigger events, message templates, and opt-in/consent capture flow were not detailed in the source notes and should be defined jointly with the client and legal/compliance stakeholders.*

18. Reporting Requirements

This section consolidates the reporting requirements referenced across Module 9 (Analytics Dashboard) and other modules.

Report	Description	Suggested Filters
Sales Report	Opportunities by stage, value, owner, and win/loss outcome.	Date range, team, owner, stage
Customer Report	Customer counts, segments, industry breakdown, activity level.	Date range, industry, segment
Revenue Report	Closed/invoiced revenue by period, product, industry, salesperson.	Date range, product, industry, salesperson
Team Performance Report	Target attainment, activity volume, and commission by salesperson/team.	Date range, team, individual
Service Report	Ticket volume, resolution time, SLA compliance, feedback score.	Date range, priority, technician
Campaign Performance Report	Sent, opened, clicked, converted per campaign.	Date range, campaign, channel

18.1 Functional Requirements

- The system shall allow all standard reports to be exported to PDF and Excel (XLSX) formats.
- The system shall allow reports to be scheduled for automatic generation and delivery via email at a configurable frequency (e.g., weekly, monthly).
- Report data visibility shall be scoped according to the requesting user's role and permissions (see Section 14).

19. Integrations

The following external integrations were identified in the client notes.

Integration	Purpose	Related Requirement(s)
WhatsApp API	Send/receive WhatsApp messages with customers.	FR-5.2, Section 17
Email API	Send/receive transactional and marketing emails.	FR-5.1, FR-7.1, Section 17
Payment Gateway	Process customer payments (e.g., against invoices/orders).	Section 2.1 (Business Model)
ERP Integration	Synchronize orders, inventory, or financial data with the client's ERP system.	Section 2.5 (Constraints)
Cloud Storage	Store customer documents and attachments.	FR-1.5
OpenAI API / LLM Provider	Power generative AI features (chat assistant, email generation, content drafting).	Section 6.10

19.1 Functional Requirements

- The system shall provide configuration screens allowing tenant administrators to connect/authorize each integration using their own credentials/API keys.
- The system shall handle integration authentication failures and expirations gracefully, notifying the administrator to re-authorize.
- The system shall log all outbound integration calls (excluding sensitive payloads) for troubleshooting and audit purposes.

ASSUMPTION: *The specific ERP system(s) and Payment Gateway provider(s) to be integrated were not named in the source notes. Integration scope (which ERP modules, which payment methods) must be confirmed per client engagement, as ERP and payment integrations are often highly client-specific.*

20. Security Requirements

The source notes did not detail security requirements explicitly; the following baseline requirements reflect standard practice for an enterprise SaaS platform handling customer and financial data, and should be reviewed against the client's compliance obligations (e.g., applicable data protection law in each operating region).

20.1 Authentication & Access Control

- The system shall require unique user credentials and support strong password policies and, for higher-privilege roles, multi-factor authentication (MFA).
- The system shall enforce role-based access control (RBAC) as defined in Section 14 on both the API and UI layers.
- The system shall enforce tenant data isolation, preventing any cross-tenant data access (see Section 9.1).

20.2 Data Protection

- All data in transit shall be encrypted using TLS 1.2 or higher (see Section 7.2).
- Sensitive data at rest (passwords, payment information, personally identifiable information) shall be encrypted using industry-standard algorithms.
- Passwords shall never be stored in plain text; they shall be hashed using a strong, salted hashing algorithm.

20.3 AI Data Handling

- Customer data transmitted to third-party AI services (e.g., OpenAI API) shall be limited to what is necessary for the specific AI feature, and shall comply with the AI provider's data-processing terms and the client's applicable data protection obligations.
- The system should provide a mechanism to exclude specific sensitive customer data from being sent to external AI providers, where required by client policy.

20.4 Audit & Compliance

- The system shall maintain an audit log of key actions (login, record creation/edit/deletion, permission changes, data export) including user, timestamp, and action.
- The system shall support data export and deletion requests to facilitate compliance with applicable data protection regulations.

ASSUMPTION: *Specific regulatory frameworks (e.g., India's Digital Personal Data Protection Act, GDPR for EU customers) were not named in the source notes. Applicable regulations should be confirmed based on the client's operating regions and customer base, and security requirements refined accordingly.*

21. Deployment Requirements

The source notes specify the technology stack (Section 8) but not deployment infrastructure details. The following baseline requirements are proposed.

- The system shall be deployed on cloud infrastructure supporting containerized or managed deployment of the Next.js frontend, NestJS/FastAPI backend services, PostgreSQL, and MongoDB.
- Separate environments shall be maintained for Development, Staging/QA, and Production, with controlled promotion between them.
- A CI/CD pipeline shall automate build, test, and deployment of the web, backend, and mobile applications.
- Database backups shall be automated on a defined schedule (e.g., daily) with a defined retention period and periodically tested restore procedure.
- The mobile application shall be published and maintained on the Google Play Store and Apple App Store, following each platform's release and update guidelines.

ASSUMPTION: *Cloud provider selection, specific CI/CD tooling, and backup retention periods were not specified in the source notes and should be finalized with the infrastructure/DevOps team.*

22. Testing Requirements

The source notes did not specify a test strategy. The following testing requirements are proposed to ensure quality across the modules described in this SRS.

Test Type	Scope
Unit Testing	Individual functions/services within backend modules (Sections 6.1–6.9) and AI service wrappers (Section 6.10).
Integration Testing	API interactions between frontend, backend, database, and third-party integrations (Section 19).
System / End-to-End Testing	Complete user workflows: Lead-to-Customer flow, Quotation flow, Service Ticket flow (Section 12).
AI Feature Validation	Accuracy and reasonableness review of AI lead scores, forecasts, and recommendations against historical data (see note below).
Performance Testing	Load and response-time testing against the targets in Section 7.1.
Security Testing	Vulnerability scanning, penetration testing, and RBAC/tenant-isolation verification (Section 20).
Usability Testing	Validation of key workflows against UI/UX requirements (Section 11) with representative users per role.
Mobile App Testing	Functional and offline-sync testing of the Sales Team App (Section 6.11) on representative Android and iOS devices.
User Acceptance Testing (UAT)	Client-led validation of all functional requirements in Section 6 against real business scenarios prior to go-live.

ASSUMPTION: *Because AI outputs (lead scores, forecasts, recommendations) are probabilistic rather than deterministic, acceptance criteria for AI features should be defined as reasonableness/accuracy thresholds validated against a labeled historical dataset, to be agreed with the client during UAT planning.*

23. Future Enhancements

The following advanced features were identified by the client as part of the long-term product vision, beyond the initial (Phase 1) release scope defined in Sections 6–22.

23.1 AI Autonomous Sales Agent

The AI shall be capable of automatically finding leads, sending outreach messages, and scheduling meetings with minimal human intervention, operating within guardrails defined by the sales team.

23.2 Voice CRM

Salespeople shall be able to update the CRM using voice commands (e.g., "Add customer meeting with ABC Industries tomorrow"), with the AI parsing the instruction and updating the relevant records automatically.

23.3 Predictive Customer Retention

The AI shall predict which customers are at risk of churn and recommend specific retention actions, extending the sentiment-analysis capability described in Section 6.10.6.

23.4 Industry Marketplace Integration

The platform shall support a connected marketplace model linking Manufacturers → Dealers → Customers, enabling multi-party transactions and visibility across the supply chain.

ASSUMPTION: *These items were described at a conceptual level in the source notes without detailed functional specification. Each should be developed into its own dedicated SRS addendum once prioritized for a future release.*

24. Risks & Limitations

Risk / Limitation	Description	Suggested Mitigation
Third-party AI dependency	Core AI features depend on the OpenAI API (or equivalent LLM provider); provider outages, pricing changes, or rate limits directly affect platform functionality.	Design AI service calls with graceful degradation, caching, and provider-abstraction to allow provider substitution.
Data privacy of AI-processed customer data	Sending customer communications and profiles to a third-party AI provider raises data protection considerations, particularly across multiple industries and possibly multiple countries.	Define a data-minimization policy for AI calls and confirm compliance requirements per client region (see Section 20.3).
Multi-industry configurability complexity	Supporting many industry-specific CRM variants from one core platform increases design and testing complexity.	Invest in a configurable data model and pipeline/stage configuration rather than hard-coded industry logic.
Integration dependency risk	WhatsApp, Email, Payment Gateway, and ERP integrations depend on external providers' availability and API stability.	Implement retry/queueing and monitoring/alerting for all integration touchpoints (Section 7.4).
Ambiguous / incomplete source requirements	The original project notes describe features at a conceptual level; several details (roles, SLAs, thresholds, pricing tiers) were not specified.	All such gaps have been documented as Assumptions throughout this SRS and must be validated with stakeholders before development begins (see Section 25.3, Open Questions).
Field connectivity for mobile app	Field sales personnel may operate in areas with poor network connectivity.	Implement offline-first data capture with background synchronization (FR-M.6).
Regulatory risk for messaging channels	WhatsApp and SMS marketing messages are subject to opt-in/consent regulations that vary by region.	Implement and enforce a consent/opt-out management mechanism (Section 17.1).

25. Appendix

25.1 Reference Source

This SRS was compiled exclusively from the client-provided project document "SmartCRM AI — AI-Powered Customer Relationship Management (CRM) Platform for Companies & Industries," covering project overview, target industries, business problems, nine application modules, nine AI features, technology stack, mobile app, business model, and future advanced features.

25.2 Technology Stack Summary

Layer	Technologies
Frontend	Next.js, React, TypeScript, Tailwind CSS
Backend	Node.js / NestJS, Python FastAPI
Database	PostgreSQL, MongoDB
AI Layer	OpenAI API, LangChain, Python ML Models, NLP Models
Integrations	WhatsApp API, Email API, Payment Gateway, ERP Integration, Cloud Storage
Mobile App	Flutter (Android & iOS)

25.3 Business Model Summary

For architectural and access-control context (see Sections 2.1 and 14), the client-described commercial model is summarized below.

Model	Detail
1. SaaS Subscription	Tiered plans: Starter, Business, Enterprise.
2. Industry-Specific CRM	Custom versions: Manufacturing CRM, Real Estate CRM, Healthcare CRM, Industrial Sales CRM.
3. AI Premium Features	Chargeable add-ons: AI analytics, sales prediction, AI assistant, automation.

25.4 Open Questions for Stakeholder Confirmation

The following questions consolidate the assumptions raised throughout this document and should be resolved with the client before or during the design phase:

- 15. Is the platform intended as a true multi-tenant SaaS product, or a separate deployment per client company? (Section 1.2)
- 16. What are the exact feature/user-limit boundaries between the Starter, Business, and Enterprise subscription plans? (Section 2.1)
- 17. What discount-approval thresholds and approval hierarchy should apply to quotations? (Section 6.4, Section 13)
- 18. What SLA durations apply to each service-ticket priority level, and do they vary by industry? (Section 6.6)
- 19. What is the required commission calculation formula, and is integration with an external payroll/HR system required? (Section 6.8)
- 20. What is the definitive role/permission matrix, and should permissions be customizable per tenant? (Section 14)

21. Which ERP system(s) and payment gateway provider(s) must be integrated, and what is the integration scope? (Section 19)
22. Which data protection regulations apply, based on the client's operating regions and customer base? (Section 20)
23. What are the target uptime SLA, expected concurrent user counts, and data volume projections used to size performance requirements? (Section 7)
24. Is a customer-facing self-service portal in scope for this phase, or is CRM access restricted to internal users only? (Section 2.3)

25.5 Document Control

This SRS should be reviewed and formally approved by the client's designated business sponsor and the project's technical lead prior to the commencement of detailed design and development. Any subsequent change to scope, modules, or requirements should be captured through a formal change request and reflected in an updated document version (see Document Version History).